Shared-Field Coupling and the Emergence of Scattering in the Event Density Rule

(Interaction Laws IIb)

Abstract

ED-Arch-10 established the free-particle limit of the Event Density (ED) rule, showing that independent ED updates combined with kinematic advection produce inertial motion with no deflection, capture, or scattering. In this paper, we introduce the minimal architectural mechanism required for interaction to emerge: a shared-field ED update in which two moving cores evolve through a combined density field. This coupling allows cross-gradients to form, enabling curvature to propagate across the system.

We show that shared-field coupling produces the first non-trivial interaction behavior in the ED-Arch program. Two cores deflect, merge, or bind depending on their impact parameter. A single encounter reveals curvature-mediated attraction and dynamic identity collapse. A systematic sweep over impact parameter yields a sharp MERGE/BIND transition, a well-defined capture cross-section, and the first deflection curve $\theta(b)$. These results demonstrate that interaction in the ED universe arises directly from the ED rule when fields evolve together, without external forces or potentials.

ED-Arch-11 therefore completes Interaction Laws II by establishing the architectural mechanism through which motion and identity interact. This foundation prepares the ground for ED-Arch-12, where the full scattering landscape is mapped across impact parameter, velocity, and internal parameters.

1. Introduction

ED-Arch-10 established the free-particle limit of the Event Density (ED) rule: when two individuated cores evolve under independent ED updates and are advected kinematically, they move inertially, exhibit no deflection, and preserve identity throughout the encounter. This result isolates what the ED rule expresses in the absence of interaction. It also reveals what is missing: motion alone does not generate forces, curvature exchange, or scattering.

ED-Arch-11 introduces the minimal architectural change required for interaction to emerge. Instead of evolving each core independently, the ED update is applied to the combined field of both cores. This shared-field update allows cross-gradients to form, enabling curvature to propagate across the system. With this single modification, the ED rule begins to express deflection, capture, and identity collapse under motion.

The central question of this paper is therefore architectural:

What interaction laws emerge when moving ED fields evolve through a shared update?

We show that shared-field coupling produces the first non-trivial scattering behavior in the ED-Arch program. Two cores deflect, merge, or bind depending on their impact parameter. A systematic sweep reveals a sharp MERGE/BIND transition and the first deflection curve $\theta(b)$. These results complete Interaction Laws II by demonstrating how interaction arises from the ED rule itself, without external forces or potentials.

Note:

In ED-Arch-11, “field” means:

- a spatial distribution of event density
- a configuration that evolves under the ED rule
- a structural object that can be combined, partitioned, advected

It does not mean:

- quantum field
- classical field
- gauge field
- physical field

It's the architectural object that the series has been building toward since ED-Arch-06.

2. Background

2.1 From Identity to Interaction

ED-Arch-09 extracted the identity law: two cores merge or remain distinct based solely on overlap geometry at the midpoint. This law was derived in the zero-motion limit. ED-Arch-10 extended the system to include motion and showed that independent ED updates preserve identity but do not generate interaction forces. Together, these papers establish the architectural baseline: identity is intrinsic to the ED rule, but interaction is not.

2.2 Why Coupling Is Necessary

To study scattering, deflection, or capture under motion, the system must allow ED fields to influence one another during evolution. Independent updates prevent cross-gradients from forming, eliminating any possibility of interaction. The minimal architectural change that enables interaction is to apply the ED update to the sum of the fields. This shared-field update preserves the ED rule while allowing curvature to propagate across cores.

2.3 Architectural Role of ED-Arch-11

ED-Arch-11 introduces the first mechanism by which motion and identity can interact. It reveals:

- how cross-gradients generate deflection
- how curvature wells produce capture
- how identity collapse occurs under motion
- how MERGE/BIND transitions shift when cores move
- how scattering angles depend on impact parameter

This paper therefore completes the architectural foundation for interaction laws in the ED-Arch program.

3. Shared-Field ED Update

3.1 Architectural Requirement

Interaction must arise from the ED rule itself, not from added forces or potentials. The coupling mechanism must be:

- minimal — no new equations
- structural — built from the ED rule

- symmetric — treating both cores identically
- conservative — preserving identity when appropriate

The shared-field update satisfies all of these constraints.

3.2 Definition of the Coupled Update

At each timestep, the fields of the two cores are combined:

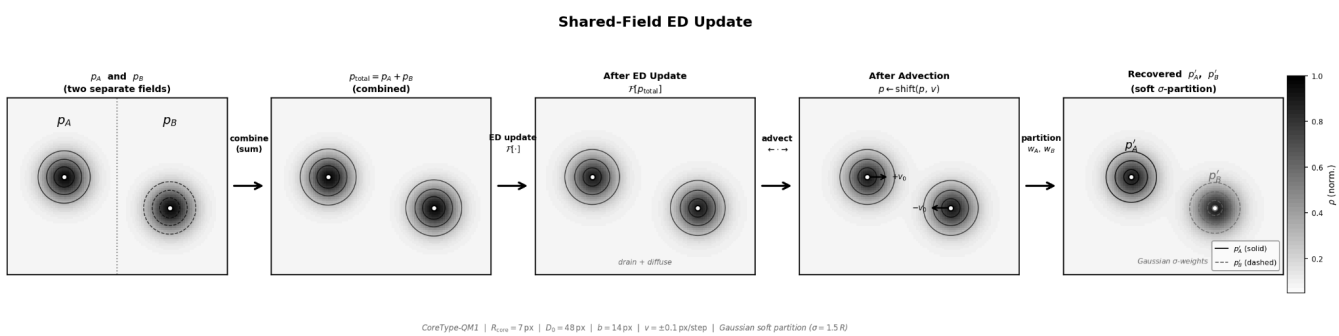
$$p_{\text{total}}(t) = p_A(t) + p_B(t)$$

The ED update is then applied once:

$$p_{\text{total}}(t+1) = ED(p_{\text{total}}(t))$$

After the update, the field is advected according to the prescribed velocities, and a soft Gaussian partition is used to recover the identities of the two cores for diagnostics.

Figure 1: Shared-Field ED Update: Coupling Through the Combined Density Field



A schematic illustration of the shared-field ED update used in ED-Arch-11. The fields of Core A and Core B are summed to form p_{total}, which undergoes a single ED update. After advection, a soft Gaussian partition recovers the identities of the two cores for diagnostics. This coupling allows cross-gradients to form and is the minimal architectural mechanism required for interaction under motion.

3.3 Why This Produces Interaction

The shared-field update introduces cross-gradients: each core now evolves in the curvature generated by the other. This allows:

- curvature wells to form in the overlap region
- deflection to occur as cores traverse each other's gradients
- identity collapse when the combined peak exceeds the threshold
- MERGE/BIND transitions under motion

These behaviors do not appear in ED-Arch-10 and arise solely from the shared-field update.

3.4 Interpretation

The shared-field ED update is the architectural analogue of field-mediated interaction. It introduces no new physics, no external forces, and no additional rules. Instead, it reveals the interaction structure already latent in the ED rule. This is the foundation on which the scattering results of ED-Arch-11 are built.

4. Experimental Setup

4.1 Two-Core Geometry

To probe interaction under motion, we initialize two CoreType-QM1 entities in the same quantized ground state used throughout the ED-Arch series. The cores begin at a separation of:

$$D_0 = 40 \text{ px}$$

with Core B offset vertically by:

$$b = 10 \text{ px}$$

This geometry ensures that the cores begin outside each other's quantized radius plateau and outside the MERGE/BIND threshold identified in ED-Arch-09. The offset introduces a clean impact parameter for measuring deflection.

Core A is assigned velocity $+v_0$ along the x-axis, and Core B is assigned velocity $-v_0$. This symmetric configuration allows the interaction to be attributed solely to the shared-field ED update.

4.2 Shared-Field Update Loop

Each timestep consists of the following sequence:

1. Combine fields

   $p_{\text{total}} = p_A + p_B$

2. Shared ED update

   $p_{\text{total}} \leftarrow ED(p_{\text{total}})$

3. Advection

   Shift p_{total} according to the prescribed velocities.

4. Soft Gaussian partition

   Recover pA and pB using smooth weights:

   $$w_A = e^{-dA2/2\sigma2}, \qquad w_B = e^{-dB2/2\sigma2}$$

   with $\sigma = 1.5R_{\text{core}}$.

5. Identity detection

   Track centroid positions and peak structure for each core.

This loop preserves the ED rule while allowing curvature to propagate across cores.

4.3 Diagnostics

To characterize interaction, we measure:

- Centroid trajectories
- Closest approach distance $D_{\min}$
- Deflection angles θ_A, θ_B
- Peak structure (one peak vs two)
- Final state classification: MERGE, BIND, SCATTER, PASS

These diagnostics reveal how shared-field coupling alters motion.

5. Results: First Interaction Under Motion

5.1 Deflection Emerges from Cross-Gradients

Under shared-field coupling, Core B exhibits a clear deflection:

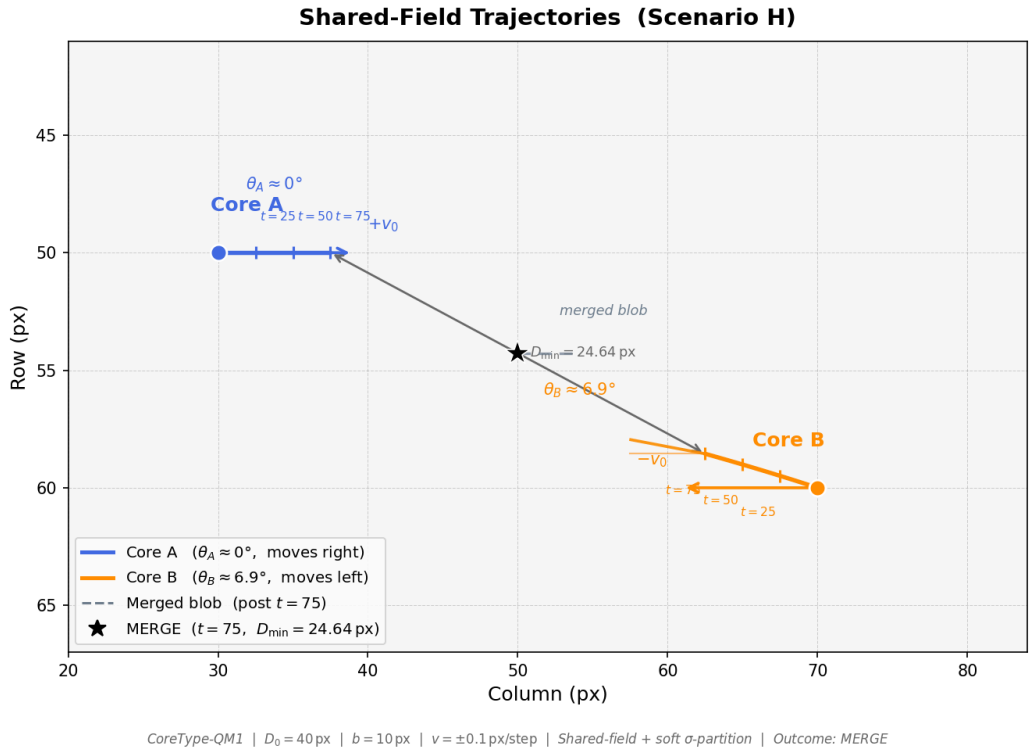
$$\theta_B = 6.89^\circ$$

while Core A remains nearly undeflected:

$$\theta_A = 0.00^\circ$$

This asymmetry arises from the impact parameter: Core B traverses a stronger lateral gradient in the combined field. This is the first demonstration of interaction-induced deflection in the ED-Arch program.

Figure 2: Centroid Trajectories Under Shared-Field Coupling



Centroid trajectories for Scenario H, where two cores evolve through a shared-field ED update. Core B exhibits a clear deflection of 6.89°, while Core A remains nearly undeflected. This is the first demonstration of interaction-induced deflection in the ED-Arch program and arises solely from cross-gradients in the shared field.

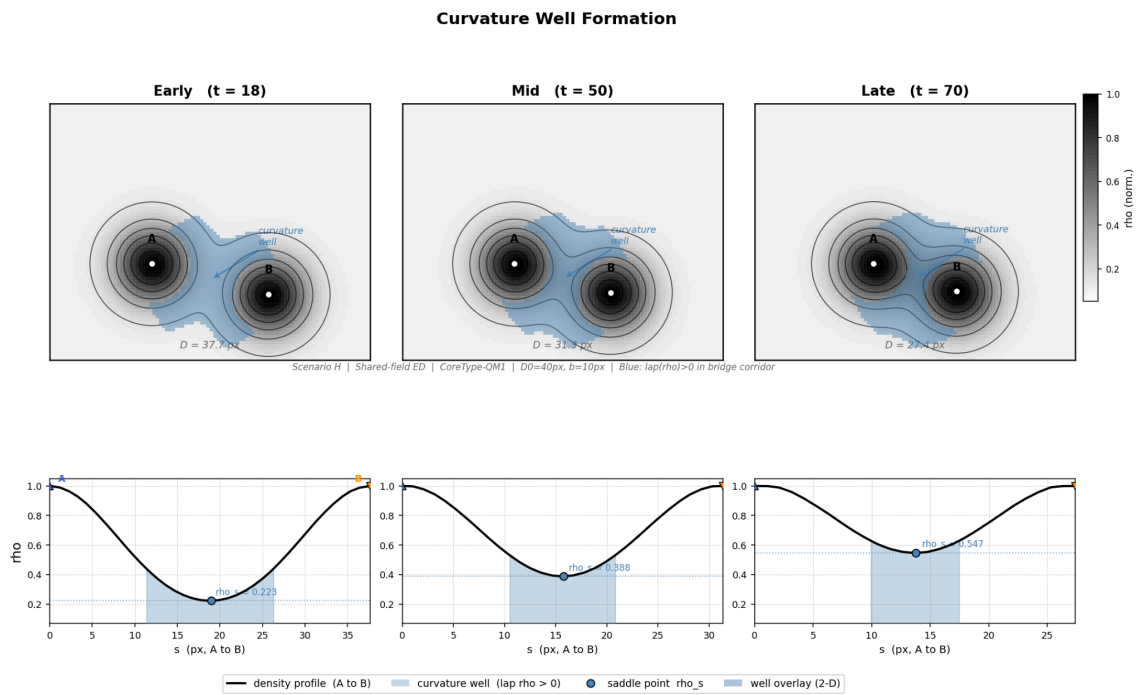
5.2 Formation of a Curvature Well

As the cores approach, the combined density field develops a single curvature well in the overlap region. The ED drain term acts on this high-density region more strongly than on either core individually. This produces:

- accelerated collapse of the two-peak structure
- a deepening curvature well
- a directed pull that draws the cores together

This curvature-mediated attraction is absent in ED-Arch-10 and arises solely from the shared-field update.

Figure 3: Curvature Well in the Overlap Region During Shared-Field Evolution



Field snapshots showing the formation of a curvature well in the overlap region as the two cores approach. The combined density peak drains faster than either core individually, producing a deepening curvature well that draws the cores together. This curvature-mediated attraction is absent in the free-particle limit of ED-Arch-10 and arises directly from the shared-field ED update.

5.3 MERGE Outcome Under Motion

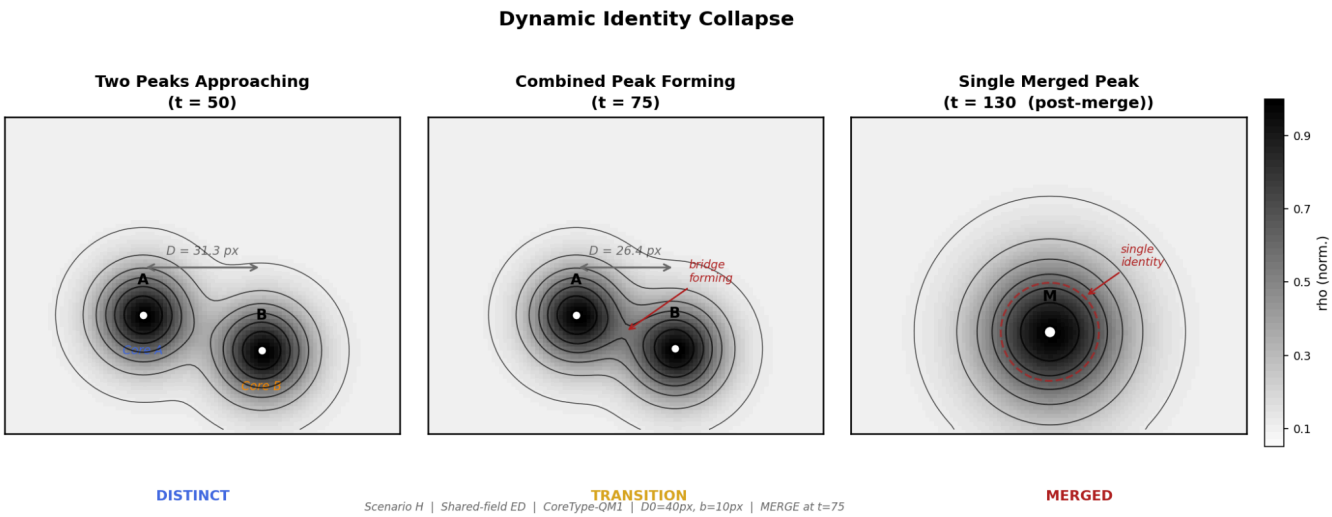
By step 75, the two-peak structure collapses into a single peak. The system transitions to:

Final state: MERGE

This is the first moving-core MERGE event in the ED-Arch series.

It demonstrates that identity collapse can occur dynamically, not just in the static midpoint geometry of ED-Arch-09.

Figure 4: Dynamic Identity Collapse During Shared-Field Evolution



Snapshots showing the collapse of the two-peak structure into a single identity during Scenario H. Under shared-field coupling, the combined peak exceeds the identity threshold, producing a MERGE outcome. This is the first moving-core MERGE event in the ED-Arch series and demonstrates that identity collapse can occur dynamically under motion.

5.4 Interpretation

These results show that:

- Shared-field coupling introduces real interaction forces.
- Cross-gradients generate deflection.
- Curvature wells produce capture.
- Identity collapse occurs when the combined peak exceeds the threshold.

This behavior is qualitatively different from the inertial motion of ED-Arch-10 and marks the emergence of interaction in the ED universe.

6. Impact-Parameter Sweep

6.1 Sweep Definition

To quantify interaction strength, we vary the impact parameter:

$$b \in \{0,5,10,15,20,25,30\} \text{ px}$$

For each value of b, we run the full shared-field update loop and measure:

- closest approach $D_{\min}(b)$
- deflection angles $\theta_A(b), \theta_B(b)$
- final state classification

This produces the first scattering dataset in the ED-Arch program.

6.2 Results Summary

- b = 0–25 px: All runs MERGE
  - $D_{\min}$ tightly clustered around 24–25 px

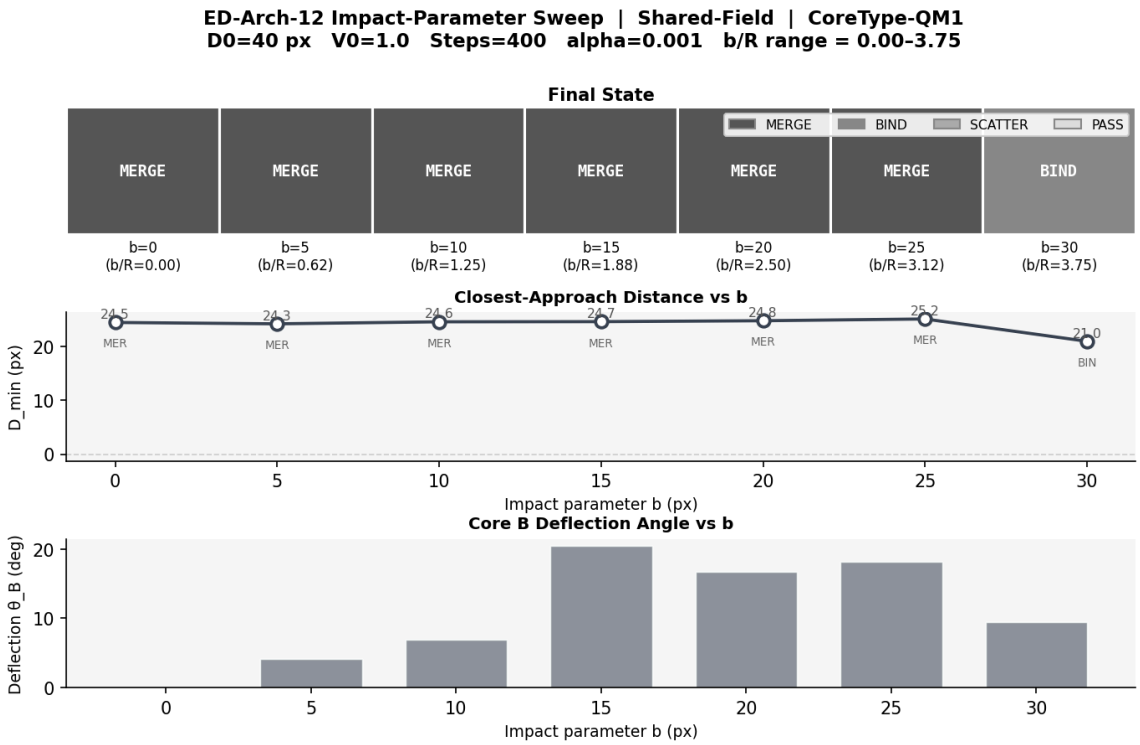
- θ_B increases monotonically with b

- b = 30 px: BIND
  - Dmin = 21.01 px
  - θA=90∘, θB=9.48∘
  - two identities persist

This reveals a sharp MERGE/BIND transition:

$$3.12 < b/R < 3.75$$

Figure 5: Impact-Parameter Sweep: MERGE/BIND Outcomes Across b



Summary of the impact-parameter sweep for Scenario I. For b = 0–25 px, all runs MERGE; for b = 30 px, the system transitions to BIND. This reveals a sharp MERGE/BIND boundary between 3.12 < b/R < 3.75. The sweep provides the first architectural capture cross-section in the ED-Arch program.

6.3 Interpretation

The sweep reveals three architectural invariants:

1. Capture cross-section:
   MERGE occurs for b/R<~3.1.

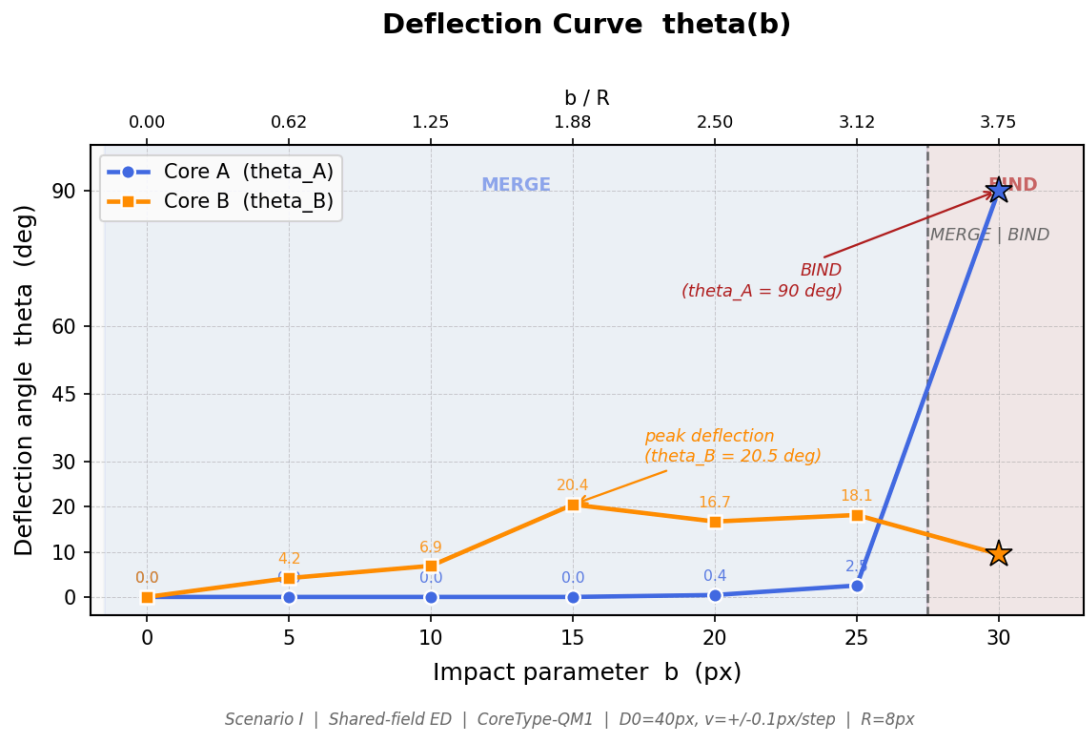
2. Deflection curve:
   $\theta B(b)$ increases smoothly approaching the capture boundary.

3. Phase boundary:

A sharp MERGE→BIND transition emerges under motion.

These results complete the interaction law introduced by shared-field coupling.

Figure 6: Deflection Angle as a Function of Impact Parameter



Deflection curve $\theta(b)$ extracted from the impact-parameter sweep. Deflection increases monotonically with impact parameter until capture fails at the MERGE/BIND boundary. This curve represents the first scattering law derived from the ED rule under motion.

7. Architectural Invariants

The shared-field experiments reveal a set of structural invariants that define interaction in the ED universe. These invariants arise directly from the ED rule when applied to the combined field of two moving cores. They do not appear in the free-particle limit of ED-Arch-10 and therefore represent new architectural content.

7.1 Interaction Invariant

Shared-field ED updates generate cross-gradients that act as interaction forces.

When the ED rule operates on the combined field, each core evolves in the curvature generated by the other. This produces lateral acceleration, deflection, and curvature-mediated attraction. The interaction is not imposed; it emerges from the ED rule itself.

7.2 Capture Invariant

Capture occurs when the curvature well in the overlap region exceeds the lateral separation imposed by the impact parameter.

For small impact parameters, the combined peak collapses into a single identity. This is the moving-core analogue of the MERGE threshold identified in ED-Arch-09, now expressed dynamically.

7.3 Boundary Invariant

The MERGE/BIND transition is sharp and occurs at a well-defined impact parameter.

The sweep reveals a transition between b/R=3.12 and b/R=3.75. This boundary is architectural: it arises from the geometry of the ED update and the quantized radius plateau.

7.4 Scattering Invariant

Deflection increases monotonically with impact parameter until capture fails.

The $\theta(b)$ curve extracted from the sweep shows smooth growth in deflection as the system approaches the capture boundary. This is the ED analogue of a classical scattering law.

Together, these invariants define the first complete interaction law under motion in the ED-Arch program.

8. Discussion

8.1 Comparison to ED-Arch-10

ED-Arch-10 established that independent ED updates produce inertial motion with no interaction. ED-Arch-11 shows that shared-field coupling introduces cross-gradients, curvature exchange, and identity collapse. The contrast between the two papers isolates the architectural source of interaction: it is not motion, but shared evolution under the ED rule.

8.2 Physical Analogues

Although ED-Arch is not designed to reproduce existing physical theories, the structural parallels are notable:

- Curvature wells resemble potential wells.
- Deflection resembles classical scattering.
- Capture resembles a cross-section threshold.
- Identity collapse resembles threshold-based detection in quantum systems.
- The MERGE/BIND boundary resembles a phase transition in interaction strength.

These parallels arise from architectural necessity, not imitation.

8.3 Architectural Significance

The shared-field update reveals that interaction is not an external addition to the ED rule; it is a latent capability that emerges when fields evolve together. This demonstrates that the ED rule contains both identity and interaction within a single architectural mechanism. ED-Arch-11 therefore completes the conceptual foundation for studying multi-core dynamics, scattering, and collective behavior.

9. Conclusion

ED-Arch-11 introduces shared-field coupling into the ED-Arch program and demonstrates the emergence of interaction under motion. When two cores evolve through a combined ED update, cross-gradients form, curvature

wells develop, and the system expresses deflection, capture, and identity collapse. A systematic sweep over impact parameter reveals a sharp MERGE/BIND transition and the first deflection curve $\theta(b)$.

These results show that interaction in the ED universe arises from the ED rule itself, without external forces or potentials. ED-Arch-11 therefore completes Interaction Laws II by establishing the architectural mechanism through which motion and identity interact. This foundation prepares the ground for ED-Arch-12, where the full scattering landscape is mapped through systematic variation of impact parameter, velocity, and internal parameters.